MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – CHAPTER TEST (SET – 6)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM:** 20

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. The midpoint of the line segment joining $(4, -2)$ and $(-2, 4)$ is: [1]

(A) $(1, 1)$ **(B)** $(2, 2)$ **(C)** $(3, 3)$ **(D)** $(0, 0)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $(1, 1)$

Given: Points $(4, -2)$ and $(-2, 4)$.

Formula Used: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

$$M = \left(\frac{4 + (-2)}{2}, \frac{-2 + 4}{2}\right) = \left(\frac{2}{2}, \frac{2}{2}\right) = \boxed{(1, 1)}$$

Q2. The line represented by $\frac{x}{4} + \frac{y}{6} = 1$ intersects x-axis and y-axis respectively at P and Q. The coordinates of the mid-point of line segment PQ are: [1]

(A) $(2, 3)$ **(B)** $(3, 2)$ **(C)** $(2, 0)$ **(D)** $(0, 3)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $(2, 3)$

Given: Line $\frac{x}{4} + \frac{y}{6} = 1$.

At x-axis $(y = 0)$: $\frac{x}{4} = 1 \implies x = 4 \implies P(4, 0)$.

At y-axis $(x = 0)$: $\frac{y}{6} = 1 \implies y = 6 \implies Q(0, 6)$.

Formula Used: Midpoint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

$$M = \left(\frac{4 + 0}{2}, \frac{0 + 6}{2}\right) = \boxed{(2, 3)}$$

Q3. If the mid-point of the line segment joining the points $(a, 4)$ and $(2, 2b)$ is $(2, 6)$, then the value of $(a + b)$ is given by: [1]

(A) 6 **(B)** 7 **(C)** 8 **(D)** 16

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 6

Given: Endpoints $(a, 4)$ and $(2, 2b)$; midpoint $(2, 6)$.

Formula Used: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Equating the abscissae and ordinates to $(2, 6)$:

$$\frac{a+2}{2} = 2 \implies a + 2 = 4 \implies a = 2$$

$$\frac{4+2b}{2} = 6 \implies 4 + 2b = 12 \implies 2b = 8 \implies b = 4$$

Adding a and b:

$$a + b = 2 + 4 = \boxed{6}$$

Q4. The mid-point of the line segment joining the points $P(-4, 5)$ and $Q(4, 6)$ lies on: [1]

(A) x-axis **(B)** y-axis **(C)** origin **(D)** neither x-axis nor y-axis

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) y-axis

Given: $P(-4, 5)$ and $Q(4, 6)$.

Formula Used: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

$$M = \left(\frac{-4+4}{2}, \frac{5+6}{2}\right) = \left(0, \frac{11}{2}\right)$$

Since the x-coordinate is 0 (while $y \neq 0$), the midpoint lies on the $\boxed{\text{y-axis}}$.

Q5. If the points $A(5, 4)$ and $B(x, 6)$ are on a circle with centre $(3, 4)$, then the value of x is: [1]

(A) 6 **(B)** 7 **(C)** 3 **(D)** 1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 3

Given: Centre $O(3, 4)$ and points $A(5, 4)$, $B(x, 6)$ lie on the circle $\implies OA = OB$ (radii).

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

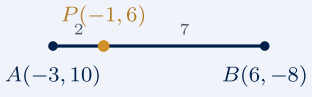
Equating $OB^2 = OA^2$:

$$(x-3)^2 + (6-4)^2 = (5-3)^2 + (4-4)^2$$

$$(x-3)^2 + 2^2 = 2^2 + 0 \implies (x-3)^2 + 4 = 4 \implies (x-3)^2 = 0 \implies \boxed{\mathbf{x = 3}}$$

Q6. Find the ratio in which the line segment joining the points $(-3, 10)$ and $(6, -8)$ is divided by $(-1, 6)$. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Setting Ratio $k : 1$ [0.5 Mark]
Given: Endpoints $A(-3, 10)$ and $B(6, -8)$, and dividing point $P(-1, 6)$. Let P divide segment AB internally in the ratio $k : 1$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\dfrac{m_1x_2 + m_2x_1}{m_1 + m_2}, \dfrac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$

Step 2: Solving from Abscissa and Verifying with Ordinate [1.5 Marks]

$$\frac{k(6) + 1(-3)}{k + 1} = -1 \implies 6k - 3 = -k - 1 \implies 7k = 2 \implies k = \frac{2}{7}$$

Checking ordinate for $m_1 : m_2 = 2 : 7$: $y = \dfrac{2(-8) + 7(10)}{2 + 7} = \dfrac{-16 + 70}{9} = \dfrac{54}{9} = 6$ (Verified).

$$\boxed{\text{Required ratio } = \mathbf{2 : 7}}$$

Q7. Find the coordinates of the point which divides the line segment joining the points $(4, -3)$ and $(8, 5)$ in the ratio $3 : 1$ internally. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Section Formula Setup [0.5 Mark]
Given: Endpoints $A(4, -3)$ and $B(8, 5)$, and internal division ratio $m_1 : m_2 = 3 : 1$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\dfrac{m_1x_2 + m_2x_1}{m_1 + m_2}, \dfrac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$

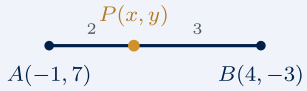
Step 2: Substitution and Evaluation [1.5 Marks]

$$x = \frac{3(8) + 1(4)}{3 + 1} = \frac{24 + 4}{4} = \frac{28}{4} = 7, \qquad y = \frac{3(5) + 1(-3)}{3 + 1} = \frac{15 - 3}{4} = \frac{12}{4} = 3$$

$$\boxed{P(x, y) = (\mathbf{7}, \mathbf{3})}$$

Q8. Find the coordinates of the point which divides the line segment joining the points $(-1, 7)$ and $(4, -3)$ in the ratio $2 : 3$ internally. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Section Formula Setup [0.5 Mark]
Given: Endpoints $A(-1, 7)$ and $B(4, -3)$, and internal division ratio $m_1 : m_2 = 2 : 3$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\dfrac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \dfrac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$

Step 2: Substitution and Evaluation [1.5 Marks]

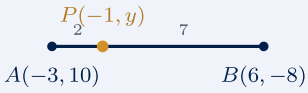
$$x = \frac{2(4) + 3(-1)}{2 + 3} = \frac{8 - 3}{5} = \frac{5}{5} = 1, \qquad y = \frac{2(-3) + 3(7)}{2 + 3} = \frac{-6 + 21}{5} = \frac{15}{5} = 3$$

$$\boxed{P(x, y) = (\mathbf{1}, \mathbf{3})}$$

SECTION C: Long Answer & Competency Based Questions MATHS MANTHAN | (9 Marks)

Q9. Find the ratio in which the point $P(-1, y)$ lying on the line segment joining points $A(-3, 10)$ and $B(6, -8)$ divides it. Also find the value of y. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Step 1: Finding the Internal Ratio [2.5 Marks]
Given: Endpoints $A(-3, 10)$ and $B(6, -8)$, and point $P(-1, y)$ on segment AB. Let P divide AB in the ratio $k : 1$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\dfrac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \dfrac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$

Equating the abscissa of P to -1:

$$\frac{k(6) + 1(-3)}{k + 1} = -1 \implies 6k - 3 = -(k + 1) \implies 6k - 3 = -k - 1$$

$$6k + k = 3 - 1 \implies 7k = 2 \implies k = \frac{2}{7} \implies \text{Ratio} = \mathbf{2 : 7}$$

Step 2: Finding the Value of y [2.5 Marks]
Substituting $m_1 = 2$, $m_2 = 7$, $y_1 = 10$, and $y_2 = -8$ into the y-coordinate formula:

$$y = \frac{2(-8) + 7(10)}{2 + 7} = \frac{-16 + 70}{9} = \frac{54}{9} = \mathbf{6}$$

$$\boxed{\text{Ratio } = \mathbf{2 : 7} \qquad \text{and} \qquad \mathbf{y = 6}}$$

Q10. Competency Case Study: A smart park security system tracks a drone patrol path between camera points $A(2,3)$ and $B(8,7)$ on a grid layout. [4]

(i) Find the straight distance between cameras A and B.
(ii) Find the coordinates of the central charging hub placed at the midpoint of AB.
(iii) If a sensor station C divides the patrol path AB in the ratio $1:2$, find the coordinates of C.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Given: Camera points $A(2,3)$ and $B(8,7)$ on the grid layout. We compute direct distance AB, midpoint hub M, and division point C dividing AB internally in ratio $1:2$.

We know the distance and internal section formulas:

Formula Used:
$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$
$$C(x,y) = \left(\frac{m_1x_2+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}\right)$$

(i) Straight Distance Between $A(2,3)$ and $B(8,7)$: [1 Mark]

$$AB = \sqrt{(8-2)^2+(7-3)^2} = \sqrt{6^2+4^2} = \sqrt{36+16} = \sqrt{52} = \mathbf{2\sqrt{13}} \text{ units}$$

(ii) Coordinates of Central Charging Hub (Midpoint M of AB): [1 Mark]

$$M = \left(\frac{2+8}{2}, \frac{3+7}{2}\right) = \left(\frac{10}{2}, \frac{10}{2}\right) = (\mathbf{5},\mathbf{5})$$

(iii) Sensor Station C Dividing AB in Ratio $1:2$: [2 Marks]

Substituting $m_1 = 1$, $m_2 = 2$, $A(2,3)$, and $B(8,7)$:

$$C = \left(\frac{1(8)+2(2)}{1+2}, \frac{1(7)+2(3)}{1+2}\right) = \left(\frac{8+4}{3}, \frac{7+6}{3}\right) = \left(\frac{12}{3}, \frac{13}{3}\right) = \left(\mathbf{4}, \mathbf{\frac{13}{3}}\right)$$

$$\boxed{AB = 2\sqrt{13} \text{ units}, \quad M = (5,5), \quad C = \left(4, \frac{13}{3}\right)}$$

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